Cooperation, Defection, and Adaptive Learning in Evolutionary Game Theory: An Extended Axelrod Tournament with a Q-Learning Agent

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Abstract

This paper runs an extended Axelrod tournament with 22 computer programs referred to as strategies. These are tested across three match lengths (100, 1,000, and 10,000 rounds), six noise levels (ε = 0.00–0.20), and six temptation payoff values (T = 3.5–6.0). The strategy ADAPTIVE was developed as a Q-learning agent using an 8-stage encoding to respond to the decisions of other strategies.

Three results stand out. First, ADAPTIVE ranks 19th of 22 at 100 rounds but 1st at 10,000 rounds, by a 17.1% margin, due to the agent needing time to learn. Second, at 10,000 rounds, ADAPTIVE wins across every noise level (ε = 0.01–0.20) and every temptation value (T = 3.5–6.0) tested, with winning margins ranging from 3.8% to 21.4%. Third, ε = 0.00 (zero noise) is the only condition where ADAPTIVE fails, finishing 15th. Without random variation in the environment, the agent is unable to determine which opponents can be exploited and ends up cooperating almost every round (93–100% cooperation) even against opponents that defect 70% of the time. Given enough rounds and any positive level of noise, within-match reinforcement learning wins unconditionally.

1. Introduction

Why do people cooperate when it would benefit them to act selfishly? This question sits at the center of economics, biology, and philosophy. In a single round of the Prisoner's Dilemma, the dominant strategy is for both players to defect (Figure 1), yet cooperation shows up constantly in labs, nature, and real life. Robert Axelrod's 1984 computer tournaments gave a well-known answer: in a repeated game, strategies that reciprocate cooperation, most famously Tit-for-Tat, beat purely selfish ones (Axelrod 2). Cooperation can survive when players interact over time.

		Player B	
		Cooperation	Non-cooperation
Player A	Cooperation	3 / 3	0 / 5
	Non-cooperation	5 / 0	1 / 1

Figure 1 The Prisoner's Dilemma

Axelrod kept three factors fixed: the number of rounds per match, the amount of noise (accidental move errors) presented, and the size of the temptation payoff. This study varies all three and tests whether a reinforcement learning agent can take advantage of that variation. Across 6 noise levels, 6 temptation values, and 3 match lengths totaling over 100,000 simulated games, this agent consistently achieved scores above those of the non-adaptive strategies, with one exception.

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2. Background

2.1 The Iterated Prisoner's Dilemma

The Prisoner's Dilemma (PD) is a two-player game where each player chooses to Cooperate (C) or Defect (D) at the same time. Payoffs are R for mutual cooperation, T for defecting against a cooperator, S for cooperating against a defector, and P for mutual defection. The constraints are T > R > P > S. This paper uses R = 3, S = 0, P = 1, and T = 5 as the baseline. Axelrod (1984) showed that Tit-for-Tat won two tournaments and remains stable when populations evolve under these payoffs.

2.2 Noise and Generous Tit-for-Tat

Molander (1985) and Nowak & Sigmund (1992) showed that TFT breaks down under noise: accidental defections cause retaliation, which causes more retaliation, and both players end up worse off. Generous Tit-for-Tat (GTFT) fixes this by randomly forgiving defections a fixed percentage of the time, making it the standard benchmark for noisy settings. In every condition we test with $\varepsilon > 0$ and 10,000 rounds, GTFT finishes second. ADAPTIVE beats it in all of those conditions by learning from payoff signals what each opponent is actually doing.

2.3 Noise in Learning Systems

Noise and learning have a complicated relationship in repeated games. While noise disrupts fixed rules, it can help learning agents by creating variation they can observe and learn from. Sutton & Barto (1998) make this point in the context of reinforcement learning: exploration is needed to

discover the value of actions you haven't done yet. Our ε = 0.00 result demonstrates this concretely in a game setting. Without noise, ADAPTIVE never gets to observe what happens when it defects against a non-retaliating opponent, so it can't learn that defection would be profitable.

2.4 Reinforcement Learning in Repeated Games

Sandholm & Crites (1996) showed that Q-learning agents can learn to cooperate in repeated social dilemmas. Most prior work has agents learning across many matches with the same opponents over time. ADAPTIVE learns within a single match, resetting its knowledge at the start of each new pairing. These results hold for settings where players meet anonymously and have no memory of past encounters with each other.

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3. Model and Methods

3.1 Payoff Matrix and Noise

Baseline payoffs: R = 3, T = 5, S = 0, P = 1. Section 4.4 varies T through 3.5, 4.0, 4.5, 5.5, and 6.0. Noise ε independently flips each player's move with probability ε each round. Section 4.3 tests ε through 0.00, 0.01, 0.03, 0.05, 0.10, and 0.20 with the baseline being ε = 0.05. All results are averaged over 10–50 independent trials per condition.

3.2 Strategy Set

Table 1 lists all 22 strategies across five behavioral types.

Strategy	Class	Behavioral Rule
Always Cooperate	Unconditional	Cooperates every round.
Always Defect	Unconditional	Defects every round.
Tit-for-Tat	Reciprocal	Cooperates first; copies opponent after.
Copy Defect	Reciprocal	Defects first; copies opponent after.
Generous TFT	Reciprocal	TFT with 20% chance to forgive defections.
Tit-for-Two-Tats	Reciprocal	Only retaliates after two defections in a row.
Grim Trigger	Retaliatory	Cooperates until first defection; defects forever after.
Pavlov	Imitative	Repeats last move if it matched opponent's; switches otherwise.

JOSS	Deceptive	TFT with 10% random defection each round.
Tester	Deceptive	Defects on round 1; exploits non-retaliators.
Prober	Deceptive	Opens D-C-C; exploits if opponent didn't retaliate.
Tranquilizer	Deceptive	Cooperates, then slowly introduces defections.
First By Davis	Delayed	Cooperates 10 rounds; plays Grim Trigger after.
Bully	Contrarian	Defects unless opponent defected last round.
Bourgeois	Stochastic	C or D with equal probability each round.
Random	Stochastic	C or D at 50% each round.
Random 70% Coop	Stochastic	Cooperates 70% of the time.
Random 30% Coop	Stochastic	Cooperates 30% of the time.
Soft Majority	Statistical	Cooperates if opponent's cooperation rate ≥ 50%.
Olive Branch	Adaptive	Cooperates when own average payoff falls below 2.0.
Defects Every Three	Periodic	Defects on rounds 0, 3, 6, …; cooperates otherwise.
ADAPTIVE	RL Agent	Q-learning; 8-state; optimistic initialization; decaying exploration.

Table 1. Strategy catalogue.

3.3 The ADAPTIVE Strategy

ADAPTIVE uses tabular Q-learning (Watkins & Dayan, 1992) within each match. It starts with an 8×2 Q-table initialized at Q(s, C) = 60 and Q(s, D) = 50, a bias toward cooperation at first, consistent with Axelrod's finding that strategies that cooperate early tend to do well. States are encoded by three features:

- My last move
- Opponent's last move
- Has opponent ever defected?

After each round, based on the reward it received, it changes the upcoming action. Because the learning rate is only 20%, it gradually learns, so it doesn't overcommit due to one good round.

At round 100, $\varepsilon \approx 3.4\%$, so ADAPTIVE is still exploring quite a bit. By round 1,000, $\varepsilon \approx 4.3\times10^{-5}$, and the agent plays an almost fixed policy. By round 10,000, $\varepsilon \approx 0$. This explains why ADAPTIVE performs so differently depending on match length.

3.4 Tournament Design

Each tournament runs a full round-robin: every strategy plays every other strategy and itself. Scores are summed across all matchups. Evolutionary simulations use a 100-agent population with fitness-proportionate selection and 5% mutation over 100 generations. Evolutionary figures use 1,000 rounds per match to show the dynamic process; the 10,000-round tournament tables represent fully converged policies.

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4. Results

4.1 Match Length: The Rank Reversal

Table 2 shows selected rankings at 100 rounds ($\varepsilon = 0.05$, $T = 5$). ADAPTIVE ranks 19th of 22, behind Grim Trigger, Bully, and Random. The top strategies are Random 30% Coop, First By Davis, and Grim Trigger, all strategies that defect early and collect payoffs before ADAPTIVE's Q-table has time to update. Axelrod's original winner, TFT, finishes 14th. At 100 rounds, the tournament rewards fast exploitation over learning.

Rank	Strategy	Total Score	Avg/Round	Notes
#1	RANDOM_30_COOP	5,298	52.96	Rapid exploiter
#2	First_By_Davis	5,281.5	52.82	Locks in defection early
#4	GRIM_TRIGGER	5,236.5	52.37	
#9	GENEROUS_TIT_FOR_TAT	5,118.5	51.19	
#14	TIT_FOR_TAT	5,021.5	50.22	Axelrod's winner
#19	**ADAPTIVE**	**4,765**	**47.65**	**Still learning**
#22	ALWAYS_COOPERATE	4,578.5	45.79	Fully exploited

Table 2. Selected rankings at 100 rounds (3-trial average, ε=0.05, T=5).

Table 3 shows the 10,000-round rankings under the same conditions (10-trial average). ADAPTIVE ends the game with the highest score of 597,585 and a margin of 87,328 points (17.1%) over

Generous TFT. Random 30% Coop, the 100-round winner, falls to 17th. Every strategy in the top six is either reciprocal or learning-based.

Rank	Strategy	Total Score	Avg/Round	Notes
#1	**ADAPTIVE**	**597,585**	**59.76**	**17.1% margin over GTFT**
#2	GENEROUS_TIT_FOR_TAT	510,258	51.03	
#3	TIT_FOR_TWO_TATS	504,102	50.41	
#5	PAVLOV	502,209	50.22	
#6	TIT_FOR_TAT	501,370	50.14	
#17	RANDOM_30_COOP	480,779	48.08	Was #1 at 100 rounds
#21	ALWAYS_COOPERATE	440,334	44.03	
#22	OLIVE_BRANCH	439,585	43.96	Most exploited

Table 3. Selected rankings at 10,000 rounds (10-trial average, ε=0.05, T=5).

4.2 Head-to-Head Results: How the Learning Works

Tables 4 and 5 show ADAPTIVE's individual matchup results at 100 and 10,000 rounds. At 100 rounds, ADAPTIVE cooperates 70–92% of the time against nearly everyone and loses 5 of 8 shown matchups. First By Davis beats it badly (91.7 vs. 376.7), as do Grim Trigger (114 vs. 334) and Random 30% Coop (112.3 vs. 379.0), all strategies that defect before the Q-table has caught up.

Opponent	ADAPTIVE Score	Opp Score	ADPT Coop%	Opp Coop%	Result
ALWAYS_COOPERATE	307.0	278.7	90.3%	96.0%	Win
TIT_FOR_TWO_TATS	301.7	276.7	88.0%	93.0%	Win
GENEROUS_TIT_FOR_TAT	290.0	293.3	92.3%	91.7%	~Tie
TIT_FOR_TAT	285.7	292.3	91.0%	89.7%	~Tie
ALWAYS_DEFECT	77.0	250.3	39.3%	4.7%	Loss
GRIM_TRIGGER	114.0	334.0	71.7%	27.7%	Loss

First_By_Davis	91.7	376.7	79.3%	22.3%	Loss
RANDOM_30_COOP	112.3	379.0	84.3%	31.0%	Loss

Table 4. ADAPTIVE head-to-head at 100 rounds (3-trial average).

At 10,000 rounds, the learned policy is visible. ADAPTIVE wins 7 of 11 matchups clearly. Against opponents that don't retaliate, cooperation rates drop to 12–22%: 20.9% against Always Cooperate (43,925 vs. 6,868), 15.7% against Bully (39,679 vs. 7,103), 22.0% against Olive Branch (43,715 vs. 7,203). Against reciprocal strategies, cooperation stays at 78–89% and scores are close to even. ADAPTIVE still loses to persistent defectors (Grim Trigger, Always Defect, First By Davis), who exploit its cooperating start before the Q-table can catch up.

Opponent	ADAPTIVE Score	Opp Score	ADPT Coop%	Opp Coop%	Result
OLIVE_BRANCH	43,715	7,203	22.0%	95.0%	Win
ALWAYS_COOPERATE	43,925	6,868	20.9%	95.0%	Win
BULLY	39,679	7,103	15.7%	80.9%	Win
DEFECTS_EVERY_THREE	31,995	11,557	24.1%	65.0%	Win
RANDOM_70_COOP	32,982	11,537	25.1%	68.0%	Win
GENEROUS_TIT_FOR_TAT	27,922	27,327	78.0%	79.2%	Win
TIT_FOR_TAT	27,568	29,537	88.9%	85.0%	~Tie
PAVLOV	27,587	29,499	88.6%	84.8%	~Tie
GRIM_TRIGGER	10,707	14,515	12.8%	5.2%	Loss
ALWAYS_DEFECT	10,764	14,194	11.9%	5.0%	Loss
First_By_Davis	10,737	14,401	12.5%	5.2%	Loss

Table 5. ADAPTIVE head-to-head at 10,000 rounds (50-trial average).

4.3 Evolutionary Dynamics

Evolutionary simulations support ADAPTIVE's advantage in population settings. Under baseline conditions (1,000 rounds per match, $\varepsilon = 0.05$, Figure 2), ADAPTIVE rises quickly, reaching about 65% of the population around generation 55, before partially declining as Tit-for-Two-Tats grows by benefiting from the cooperative environment ADAPTIVE creates. This 'cooperative umbrella' effect,

where ADAPTIVE's dominance helps competing strategies grow enough to push back, is a known result in evolutionary game theory. Figure 3 shows ε = 0.20: ADAPTIVE still leads throughout, but with more variation and a smaller peak (62%), as stronger noise hurts the learning signal.

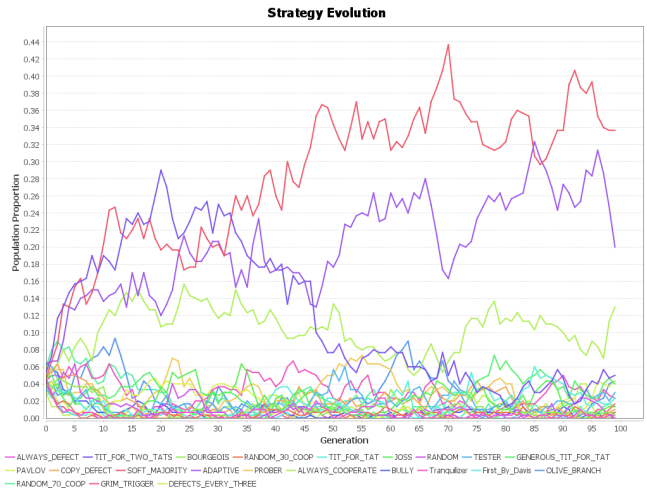

Figure 2

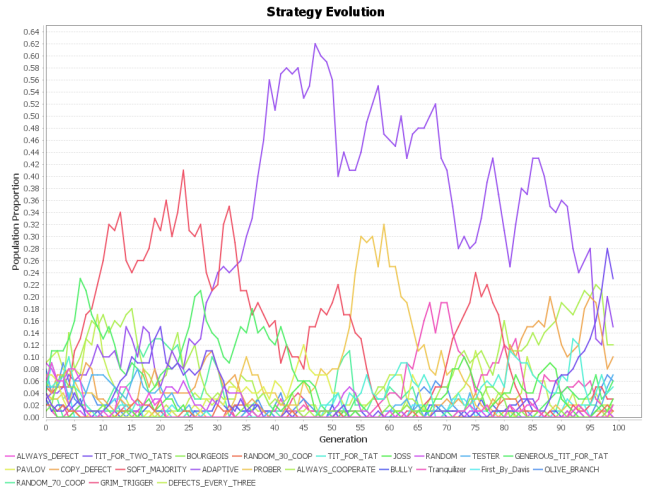

Figure 3

Figure 4 shows ε = 0.00. No strategy takes over; the population stays spread across all 100 generations. ADAPTIVE never rises above baseline, consistent with its 15th-place tournament finish at ε = 0.00 (discussed in Section 4.4). Figures 5 and 6 show T = 3.5 and T = 6.0, illustrating the payoff sensitivity covered in Section 4.5.

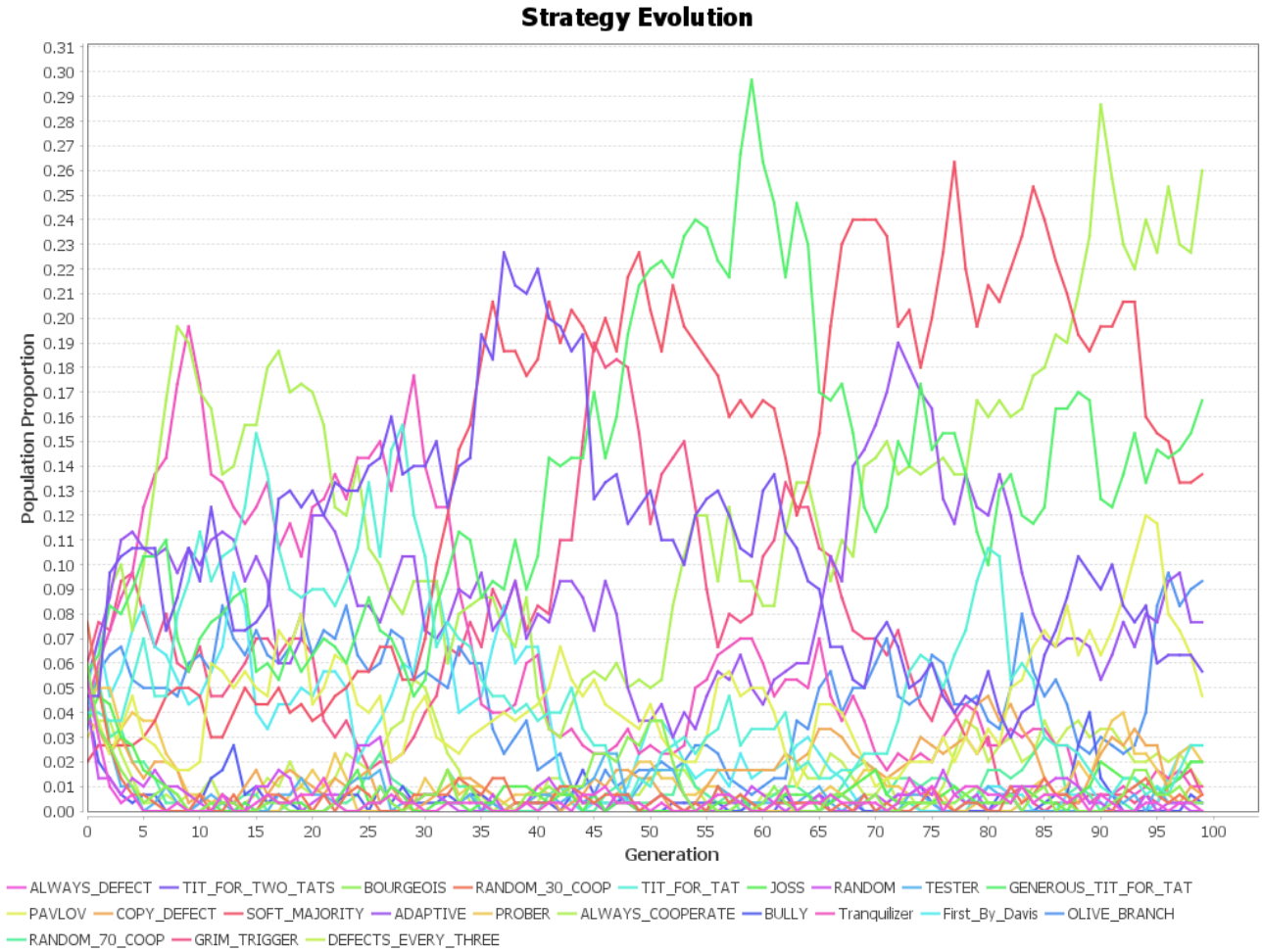

Figure 4

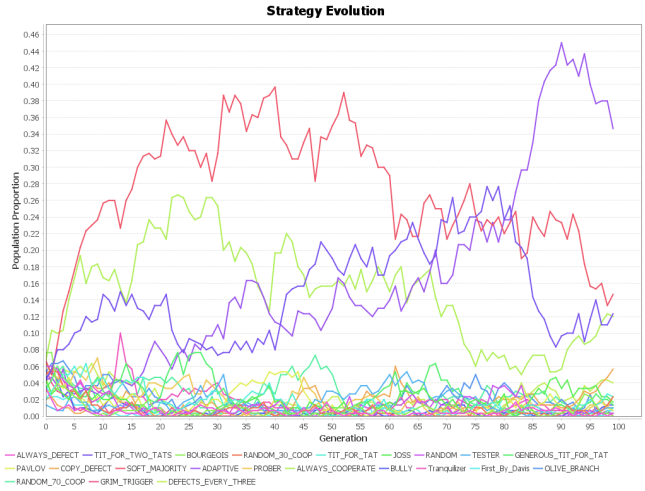

Figure 5

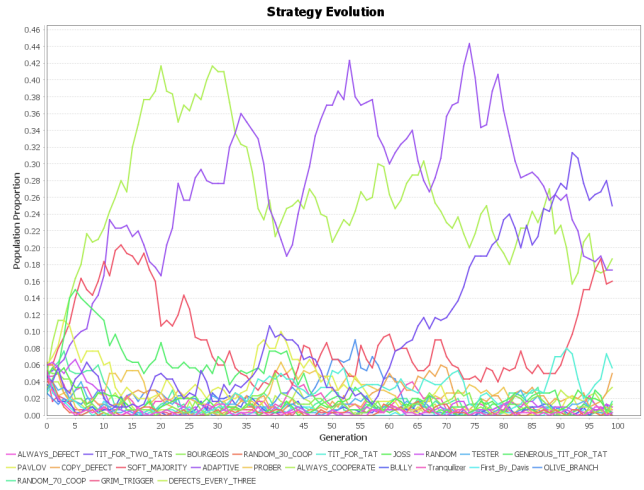

Figure 6

4.4 Noise Sensitivity: Where ADAPTIVE Dominates and Where It Doesn't

Table 6 summarizes ADAPTIVE's results across all six noise levels at 10,000 rounds. ADAPTIVE wins at every noise level above zero. At ε = 0.01, the margin is already +6.9%; it peaks at +17.1% (ε = 0.05) and stays positive even at ε = 0.20 (+3.8%). The only exception is ε = 0.00, where ADAPTIVE finishes 15th, 93,263 points behind Grim Trigger.

Noise (ε)	ADAPTIVE Rank	ADAPTIVE Score	Tournament Leader	Margin	Interpretation
0.00	#15	497,441	Grim Trigger (#1: 590,704)	−93,263	ADAPTIVE over-cooperates; noise required for learning
0.01	#1	566,555	GTFT (#2: 530,059)	+36,496 (+6.9%)	First noise level where ADAPTIVE wins
0.03	#1	600,542	GTFT (#2: 519,338)	+81,204 (+15.6%)	
0.05	#1	597,585	GTFT (#2: 510,258)	+87,328 (+17.1%)	Baseline; peak margin
0.10	#1	569,804	Pavlov (#2: 504,467)	+65,337 (+13.0%)	
0.20	#1	537,615	First_By_Davis (#2: 518,148)	+19,467 (+3.8%)	Smallest margin; high noise disrupts policy

Table 6. ADAPTIVE tournament performance across noise levels at 10,000 rounds (10-trial average).

Table 7 shows ADAPTIVE's individual matchups at ε = 0.00 and 10,000 rounds. ADAPTIVE cooperates 93–100% of the time against virtually everyone, including opponents that defect constantly. Against Random 30% Coop, ADAPTIVE cooperates 93.5% of the time and loses 9,857 to 41,573. Against Always Defect, ADAPTIVE cooperates half the time. Against Grim Trigger, 57%. In 0% noise, ADAPTIVE cooperates almost unconditionally due to it getting punished and trying to make up for it.

Opponent	ADAPTIVE Score	Opp Score	ADPT Coop%	Opp Coop%	Result
SOFT_MAJORITY	30,005	29,992	99.97%	100.00%	~Tie
TIT_FOR_TAT	29,833	29,832	98.32%	98.32%	~Tie

PAVLOV	29,996	29,996	99.96%	99.96%	~Tie
GENEROUS_TIT_FOR_TAT	29,999	29,997	99.98%	99.98%	~Tie
ALWAYS_COOPERATE	30,407	29,390	97.97%	100.00%	~Tie
DEFECTS_EVERY_THREE	21,001	34,994	94.65%	66.66%	Loss
RANDOM_70_COOP	21,536	34,963	96.88%	70.02%	Loss
BOURGEOIS	15,769	38,230	94.96%	50.04%	Loss
RANDOM	15,571	38,609	95.97%	49.89%	Loss
BULLY	12,659	31,136	68.48%	31.52%	Loss
RANDOM_30_COOP	9,857	41,573	93.46%	30.03%	Loss
GRIM_TRIGGER	5,025	32,431	57.34%	2.52%	Loss
ALWAYS_DEFECT	4,972	30,110	50.28%	0.00%	Loss
First_By_Davis	5,699	33,686	62.45%	6.47%	Loss

Table 7. ADAPTIVE head-to-head at ε=0.00, 10,000 rounds.

The cause is a failure of exploration in a noise-free environment. Without environmental noise, the only defections ADAPTIVE tries against cooperative opponents come from its own exploration steps, which are rare and declining. When ADAPTIVE does explore a defection, the opponent retaliates, and in a noise-free world, that retaliation is permanent or long-lasting (as with Grim Trigger). The Q-table update then lowers the value of defection in that state. Because ADAPTIVE weighs future payoffs heavily ($\gamma = 0.95$), the long-run cost of triggering retaliation outweighs the short-run gain from defecting, so Q(s, D) ends up below Q(s, C) in almost every state. The agent gets stuck cooperating with everyone and has no way out without some external disruption.

Noise solves this problem by producing accidental defections that opponents can't punish (since they were unintended). ADAPTIVE can then observe the payoff from defecting without triggering lasting retaliation, slowly raising the Q-value for defection against opponents who don't fight back. At $\varepsilon = 0.01$, this is already enough: ADAPTIVE ranks 1st with a +6.9% margin.

4.5 Temptation Sensitivity: ADAPTIVE Wins at Every T

Table 8 shows ADAPTIVE's results across all six temptation values at 10,000 rounds ($\varepsilon = 0.05$). ADAPTIVE ranks first in every condition. Winning margins range from 11.6% (T = 4.0) to 21.4% (T = 6.0). Unlike the noise condition, there is no temptation value where ADAPTIVE fails, with the only variation being how large the advantage is.

T	ADAPTIVE Rank	ADAPTIVE Score	2nd Place	Margin	Notes
3.5	#1	597,747	GTFT (#2: 512,592)	+85,155 (+16.6%)	
4.0	#1	545,207	TFT2T (#2: 488,617)	+56,590 (+11.6%)	Smallest margin
4.5	#1	597,106	GTFT (#2: 512,092)	+85,014 (+16.6%)	
5.0 (baseline)	#1	597,405	GTFT (#2: 512,299)	+85,106 (+16.6%)	
5.5	#1	597,288	GTFT (#2: 512,518)	+84,770 (+16.6%)	
6.0	#1	661,487	First_By_Davis (#2: 544,958)	+116,529 (+21.4%)	Largest margin

Table 8. ADAPTIVE tournament performance across temptation values at 10,000 rounds (10-trial average, ε=0.05).

At T = 4.0 (Table 9), ADAPTIVE still wins but with its smallest margin. When the temptation payoff is only slightly above the mutual cooperation reward (T = 4.0 vs. R = 3.0), there is less to gain from identifying and exploiting opponents who won't fight back. TFT2T finishes second, benefiting from the generally cooperative field ADAPTIVE creates. Even so, ADAPTIVE's ability to cooperate at high rates with reciprocating opponents (~93%) while still extracting some payoff from non-retaliators puts it at the top.

Rank	Strategy	Total Score	Avg/Round	Notes
#1	**ADAPTIVE**	**545,207**	**54.52**	
#2	TIT_FOR_TWO_TATS	488,617	48.86	Benefits from a high-cooperation environment
#3	GENEROUS_TIT_FOR_TAT	482,763	48.28	
#4	SOFT_MAJORITY	478,783	47.88	
#5	TIT_FOR_TAT	463,409	46.34	
#21	First_By_Davis	414,210	41.42	

#22	ALWAYS_DEFECT	413,988	41.40	Defection is least valuable at low T

Table 9. Selected rankings at T=4.0, 10,000 rounds.

At T = 6.0 (Table 10), ADAPTIVE achieves its best result: 661,487 points, a 21.4% margin over First_By_Davis. Against Always Cooperate, ADAPTIVE scores 52,143 while cooperating only 18.4% of the time; against Olive Branch, 52,483 (17.3%); against Bully, 47,680 (14.4%). The high temptation payoff makes each successful exploitation round worth more, which magnifies the returns from ADAPTIVE's learned defection policy. Defection-heavy strategies (Always Defect, Grim Trigger, First_By_Davis) also rank well at T = 6.0, but ADAPTIVE's advantage over them comes from earning R = 3 per round against reciprocal strategies rather than the mutual defection payoff of P = 1.

Rank	Strategy	Total Score	Avg/Round	Notes
#1	**ADAPTIVE**	**661,487**	**66.15**	**21.4% margin; largest in all tests**
#2	First_By_Davis	544,958	54.50	
#3	GRIM_TRIGGER	544,758	54.48	
#4	ALWAYS_DEFECT	544,552	54.46	
#5	GENEROUS_TIT_FOR_TAT	543,544	54.35	
#7	TIT_FOR_TAT	542,621	54.26	
#21	ALWAYS_COOPERATE	446,919	44.69	
#22	OLIVE_BRANCH	446,838	44.68	Most exploited at T=6

Table 10. Selected rankings at T=6.0, 10,000 rounds.

Table 11 shows ADAPTIVE's matchups at ε = 0.20, the highest noise level where ADAPTIVE still wins. Even with heavy signal disruption, ADAPTIVE correctly reads most opponents: 23.6% cooperation against Olive Branch (37,731 vs. 9,558), 23.8% against Always Cooperate (37,728 vs. 9,621), 53.3% against Generous TFT (24,821 vs. 22,716). The policy is noisier than at ε = 0.05(cooperation against TFT and Pavlov drops to 60–61% instead of 88–89%)but ADAPTIVE still broadly gets opponents right.

Opponent	ADAPTIVE	Opp	ADPT	Opp	Resul

	Score	Score	Coop%	Coop%	t
OLIVE_BRANCH	37,731	9,558	23.6%	79.9%	Win
ALWAYS_COOPERATE	37,728	9,621	23.8%	80.0%	Win
BULLY	33,649	10,036	20.5%	67.7%	Win
DEFECTS_EVERY_THREE	30,140	12,186	24.1%	60.0%	Win
RANDOM_70_COOP	30,633	12,459	25.6%	62.0%	Win
BOURGEOIS	26,060	14,085	25.9%	49.9%	Win
GENEROUS_TIT_FOR_TAT	24,821	22,716	53.3%	57.5%	Win
TIT_FOR_TAT	23,369	25,510	60.7%	56.4%	~Tie
PAVLOV	23,440	25,639	61.3%	56.9%	~Tie
SOFT_MAJORITY	15,773	17,850	27.1%	22.9%	~Tie
GRIM_TRIGGER	14,906	17,844	26.0%	20.1%	Loss
ALWAYS_DEFECT	14,867	17,800	25.8%	19.9%	Loss
First_By_Davis	14,913	17,782	25.8%	20.0%	Loss

Table 11. ADAPTIVE head-to-head at ε=0.20, 10,000 rounds.

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5. Discussion

5.1 The Core Mechanism

Every major result in this paper comes from the same place: ADAPTIVE's Q-learning successfully figures out what each opponent is doing and adjusts, but only when two conditions are met: enough rounds to converge and non-zero noise to explore. At 10,000 rounds with ε > 0, both conditions are satisfied across all tested parameters. At 100 rounds, the convergence condition fails. At ε = 0.00, the noise condition fails. Each failure has a clear, specific cause in the Q-learning process.

The ε = 0.00 finding is counterintuitive. You might expect that less noise would make it easier to classify opponents. However, removing noise takes away the only mechanism by which ADAPTIVE can safely test whether an opponent will retaliate. In a noise-free environment, any defection ADAPTIVE makes triggers a response from reciprocal strategies, making defection look

unprofitable in the Q-table even against opponents where sustained defection would actually pay well. What looks like a benefit (a noise-free game) turns out to remove the variation that makes learning possible.

5.2 Long-Run Dominance and What It Means for Economic Design

The consistency of ADAPTIVE's long-run results is the strongest finding here. It doesn't perform well under some conditions and poorly under others; it ranks first under every noise level and every temptation value tested, as long as the match is long enough. The practical implication is direct: the value of using an adaptive, learning-based approach in a repeated interaction grows with how long that interaction lasts, and it holds up across a wide range of environmental conditions.

For economic institution design, this means that long-term relationships aren't just environments where cooperation is possible. They are environments where learning-based strategies unconditionally outperform every fixed rule, regardless of the exact payoff structure or error rate, as long as some minimal level of noise is present. The $\varepsilon = 0.00$ exception also implies that perfectly monitored, zero-noise enforcement environments may inadvertently disable adaptive cooperation by removing the exploratory variation that learning-based agents need to function.

5.3 Persistent Defectors Remain a Weakness

Across all conditions, ADAPTIVE continues to lose against Grim Trigger, Always Defect, and First By Davis, strategies that defect immediately and keep defecting. This loss is contained: at 10,000 rounds with $\varepsilon = 0.05$, ADAPTIVE scores 10,707 against Grim Trigger vs. 14,515 for Grim Trigger, but that's just one of 22 matchups. Because Grim Trigger also scores poorly against most other strategies, its strong individual result against ADAPTIVE doesn't translate to a tournament win. The 'reset probe' mechanism (trying cooperation again after a period of mutual defection to check whether the opponent would respond) would reduce this weakness and is a logical direction for future work; however, with many of the strategies, it loses to those that wouldn't make a difference. No matter what, they will always defect, so cooperation will never be reached; however, against other strategies not mentioned, this would most likely help.

5.4 The 1,000-Round Convergence Window

At 1,000 rounds per match, ADAPTIVE's performance depends on the condition: it wins at $T = 3.5$, $T \geq 5.5$, and $\varepsilon \in \{0.03, 0.05, 0.10, 0.20\}$, but ranks lower at $T = 4.0$–4.5 and $\varepsilon \in \{0.00, 0.01\}$. This reflects where ADAPTIVE sits in its convergence window. At 1,000 rounds, $\varepsilon \approx 4.3\times10^{-5}$, and the policy is nearly fixed, but the Q-table has had fewer updates to refine its read of each opponent. At 10,000 rounds, those residual effects disappear. The 1,000-round results tell us how much time different conditions require before ADAPTIVE reaches full performance, not that the strategy has fundamental limits.

Evolutionary simulations at 1,000 rounds show ADAPTIVE's population advantage but also the cooperative umbrella effect: ADAPTIVE's success creates a cooperative environment that benefits forgiving strategies like TFT2T and GTFT, which then compete with ADAPTIVE for population

share. This aligns with known results on the instability of cooperation-enabling strategies in finite populations (Nowak & May, 1992) and suggests that evolutionary stability may need separate analysis from tournament dominance.

5.5 Limitations

Within-match learning only. ADAPTIVE resets its Q-table at the start of each new match. Retaining memory across matches could speed up convergence and possibly extend dominance to shorter match lengths.

Small state space. Eight states encoding three binary features is a strong simplification. Richer encodings (round number, mutual cooperation streaks, etc.) could help ADAPTIVE learn faster and do better against persistent defectors.

Hyperparameters were not tuned. $\alpha = 0.20$, $\gamma = 0.95$, and $\varepsilon_0 = 0.05$ were fixed without systematic search. The exact noise and match-length thresholds for dominance are sensitive to these values.

Tournament dominance doesn't guarantee evolutionary stability. Tournament rankings don't imply that a strategy is stable against invasion in a co-evolutionary population. Future work should test whether ADAPTIVE's dominance holds under those conditions.

---

6. Conclusion

Given 10,000 rounds, ADAPTIVE ranks first in every condition tested except $\varepsilon = 0.00$ (zero noise), where it finishes 15th. The $\varepsilon = 0.00$ exception is not a failure of the strategy but a failure of the learning condition. Without environmental noise, Q-learning can't discover that defecting against non-retaliating opponents is profitable, and the agent converges to near-universal cooperation (93–100%) even against opponents that exploit it heavily.

Three results stand out for economic theory. First, interaction length is the main driver of learning-based strategy success: 100 rounds gives 19th place, 10,000 rounds gives 1st place, and the individual matchup data shows exactly why. Second, within-match reinforcement learning is universally dominant at long horizons. The result holds across a wide range of noise levels ($\varepsilon = 0.01$–0.20) and temptation values ($T = 3.5$–6.0). Third, noise is a prerequisite for learning: perfectly monitored environments disable adaptive cooperation by removing the variation that Q-learning agents need.

For institutional design, the takeaway is that long-horizon relationships (supply chains, employment contracts, regulatory oversight, long-term agreements) are not just settings where cooperation can emerge. They are settings where learning-based adaptive agents become unconditionally dominant. Short-horizon markets favor fast exploiters; zero-noise environments may disable adaptive learners. Both the length of the relationship and the level of implementation noise should be treated as first-order considerations when designing repeated-interaction institutions.

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Appendix A: ADAPTIVE Algorithm Pseudocode

Algorithm: ADAPTIVE Agent

```
Initialization (start of each match):
  for all s in {0,...,7}:
    Q[s][C] <- 60.0   // Biases toward cooperation early
    Q[s][D] <- 50.0
  explorationRate <- 0.05

State encoding (3 binary features -> 8 states):
  state <- (lastMove==D ? 4 : 0)            // Bit 4
```

```
        + (oppLastMove==D ? 2 : 0)          // Bit 2
        + (oppEverDefected ? 1 : 0)         // Bit 1

play() -> action:
  s <- getCurrentState()
  if rand() < explorationRate:
    return randomChoice({C, D})
  else:
    return argmax_a Q[s][a]

updateAfterRound(myMove, reward, oppMove):
  prevState <- getCurrentState()
  a <- (myMove == C) ? 0 : 1
  update lastMove, lastOpponentMove, oppEverDefected
  newState <- getCurrentState()
  Q[prevState][a] <- Q[prevState][a]
      + 0.20 * (reward + 0.95 * max_{a'} Q[newState][a'] - Q[prevState][a])
  explorationRate <- explorationRate * 0.99

Key property: explorationRate at round N = 0.05 * 0.99^N
  Round 100   -> epsilon ~= 0.034    (still partly stochastic)
  Round 1,000 -> epsilon ~= 4.3e-5   (near-deterministic)
  Round 10,000-> epsilon ~= 0        (fully deterministic)
```